Probability

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Contents

Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ is a collection of subsets of Ω. We call $\mathcal{F}$ a *σ-algebra* if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^{\mathrm{c}} \in \mathcal{F}$;
- if $(A_n)_{n\in\mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ-algebra on Ω. A function

$$\mathbb{P} : \mathcal{F} \to [0,1]$$

is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;
- for any countable disjoint collection $(A_n)_{n\in\mathbb{N}}$ in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A.

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ-algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω.

> **Remark.** When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω. However, this framework will become useful when considering uncountable Ω.

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;
- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4

- *Rolling a fair die.* Then $\Omega = \{1,2,3,4,5,6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and
$$\mathbb{P}(A) = \frac{\mid A \mid}{6}$$
since all outcomes are equally likely.
- $\Omega = \{\omega_1, \dots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.
- *Picking balls from a bag.* Suppose we have n balls labelled $\{1, \dots, n\}$, and we pick $k \leq n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \dots, n\}, \mid A \mid= k\}$, so $\mid \Omega \mid= \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.
- *A deck of 52 cards.* Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $\mid \Omega \mid= 52!$.

  What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.
- *Largest digit.* Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \dots, 9\}^n$, $\mid \Omega \mid= 10^n$.

  Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then
$$\begin{aligned} \mathbb{P}(A_k) &= \frac{(k+1)^n}{10^n} \\ \implies \mathbb{P}(B_k) = \mathbb{P}(A_k \setminus A_{k-1}) &= \frac{(k+1)^n - k^n}{10^n} \end{aligned}$$
- *Birthday problem.* There are n people. What is the probability that at least 2 of them share the same birthday?

  Assume each birthday is equally likely to be one of $\{1, \dots, 365\}$. Then $\Omega = \{1, \dots, 365\}^n$, $\mid \Omega \mid= 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

  Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^c = \{\text{all birthdays different}\}$.
$$\begin{aligned} \mathbb{P}(A^c) &= \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \\ \implies \mathbb{P}(A) &= 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \end{aligned}$$
  Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $\mid \Omega \mid= n$. We want to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $\mid \Omega_i \mid= n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$
\begin{aligned}
M &= \binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdots \binom{n-(n_1+\cdots+n_{k-1})}{n_k} \\
&= \frac{n!}{n_1! n_2! \cdots n_k!} \\
&= \binom{n}{n_1, n_2, \ldots, n_k}
\end{aligned}
$$

This is the *multinomial coefficient.*

2. Suppose $f : \{1, \ldots, k\} \to \{1, \ldots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

   How many strictly increasing $f : \{1, \ldots, k\} \to \{1, \ldots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.

   What about increasing functions? Let's define a bijection

   $$\{f : \{1, \ldots, k\} \to \{1, \ldots, n\} \text{ inc.}\} \to \{f : \{1, \ldots, k\} \to \{1, n+k-1\} \text{ strictly inc.}\}$$

   which takes f to g with $g(i) = f(i) + i - 1$. Hence we have $\binom{n+k-1}{k}$ such functions.

Definition 1.5 (Asymptotic notation)**.** If (a_n) and (b_n) are sequences, we write $a_n \sim b_n$ if

$$\lim_{n \to \infty} \frac{a_n}{b_n} = 1$$

§1.2 Stirling's formula

Theorem 1.6 (Stirling)

$$n! \sim n^n \sqrt{2\pi n} e^{-n}$$

as $n \to \infty$.

Weaker statement. We shall first show a weaker statement, that $\log n! \sim n \log n$.

Define $l_n = \log n! = \log 2 + \cdots + \log n$. We have

$$\log \lfloor x \rfloor \leq \log x \leq \log \lfloor x + 1 \rfloor$$

Integrating from 1 to n gives

$$
\begin{aligned}
\sum_{k=1}^{n-1} \log k \leq \int_1^n &\log x \, dx \leq \sum_{k=1}^{n} \log k \\
\implies l_{n-1} &\leq n \log n - n + 1 \leq l_n \\
\implies n \log n - n + 1 &\leq l_n \leq (n+1) \log(n+1) - (n+1) + 1
\end{aligned}
$$

Now the result follows by dividing through by $n \log n$ and taking the limit. ■

Non-examinable proof of Stirling. For every twice-differentiable $f : \mathbb{R} \to \mathbb{R}$, for all $a < b$, we have

$$\int_a^b f(x)\ dx = \frac{f(a) + f(b)}{2}(b-a) - \frac{1}{2}\int_a^b (x-a)(b-x)f''(x)\ dx$$

obtained by integrating twice by parts. Put $f = \log$, $a = k$, $b = k+1$ for $k \in \mathbb{N}$, to get

$$\int_k^{k+1} \log x\ dx = \frac{\log k + \log(k+1)}{2} + \frac{1}{2}\int_k^{k+1} \frac{(x-k)(k+1-x)}{x^2}\ dx$$

Summing from $k = 1$ to $n-1$, we have

$$\int_1^n \log x\ dx = \frac{\log(n-1)! + \log n!}{2} + \sum_{k=1}^{n-1} a_k$$

where $a_k = \frac{1}{2}\int_0^1 \frac{x(1-x)}{(x+k)^2}\ dx$. Hence

$$n \log n - n + 1 = \log n! - \frac{\log n}{2} + \sum_{k=1}^{n-1} a_k$$

and so

$$\log n! = n \log n - n + 1 + \frac{\log n}{2} - \sum_{k=1}^{n-1} a_k$$
$$\implies n! = n^n \cdot e^{-n} \cdot \sqrt{n} \cdot \exp\left(1 - \sum_{k=1}^{n-1} a_k\right)$$

Let us now consider

$$\begin{aligned} a_k &= \frac{1}{2}\int_0^1 \frac{x(1-x)}{(x+k)^2}\ dx \\ &\le \frac{1}{2}\int_0^1 \frac{1}{2k^2}\int_0^1 x(1-x)\ dx \\ &= \frac{1}{12k^2} \end{aligned}$$

and so $\sum_{k=1}^{\infty} a_k < \infty$. Now put

$$A = \exp\left(1 - \sum_{k=1}^{\infty} a_k\right)$$
$$\implies n! = n^n \cdot e^{-n} \cdot \sqrt{n} \cdot A \cdot \underbrace{\exp\left(\sum_{k=n}^{\infty} a_k\right)}_{\to 1 \text{ as } n \to \infty}$$
$$\implies \frac{n!}{n^n e^{-n} \sqrt{n}} \to A \text{ as } n \to \infty$$

It remains to show that $A = \sqrt{2\pi}$. Consider

$$2^{-2n}\binom{2n}{n} = 2^{-2n}\frac{(2n)!}{(n!)^2}$$

$$\sim \frac{2^{-2n}(2n)^{2n}e^{-2n}\sqrt{2n}A}{n^{2n}e^{-2n}nA^2}$$
$$= \frac{\sqrt{2}}{A\sqrt{n}}$$

Now, we shall use a different method to show

$$2^{-2n}\binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \text{ as } n \to \infty$$

Consider

$$I_n = \int_0^{\pi/2} \cos^n\theta \, d\theta$$

We have $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and, by integrating by parts, we have

$$I_n = \frac{n-1}{n} I_{n-2} \tag{$*$}$$

so that

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \cdots = \frac{(2n-1)(2n-3)\cdots 3\cdot 1}{(2n)(2n-2)\cdots 2}\cdot\frac{\pi}{2} \\ &= \frac{2n(2n-1)(2n-2)\cdots 3\cdot 2\cdot 1}{2n\cdot 2n\cdot(2n-2)\cdot(2n-2)\cdots 2\cdot 2}\cdot\frac{\pi}{2} \\ &= \frac{(2n)!}{2^{2n}(n!)^2}\cdot\frac{\pi}{2} \\ &= \frac{\pi}{2}\cdot 2^{-2n}\binom{2n}{n} \end{aligned}$$

and we can similarly obtain

$$\begin{aligned} I_{2n+1} &= \frac{2n\cdots 4\cdot 2}{(2n+1)\cdots 3\cdot 1}\cdot 1 \\ &= \frac{1}{2n+1}\left(2^{-2n}\binom{2n}{n}\right)^{-1} \end{aligned}$$

Now, we want to show that $\frac{I_{2n}}{I_{2n+1}} \to 1$ as $n \to \infty$. By ($*$), we have

$$\frac{I_n}{I_{n-2}} \to 1 \text{ as } n \to \infty$$

But $I_n = \int_0^{\pi/2} \cos^n\theta \, d\theta$, so I_n is decreasing in n. This gives

$$\frac{I_{2n}}{I_{2n-2}} \le \frac{I_{2n}}{I_{2n+1}} \le \frac{I_{2n-1}}{I_{2n+1}}$$

which shows the result by sandwich. Hence, we get, as $n \to \infty$,

$$\frac{\frac{\pi}{2}\cdot 2^{-2n}\cdot\binom{2n}{n}}{\frac{1}{2n+1}\cdot\left(2^{-2n}\binom{2n}{n}\right)^{-1}} \to 1$$
$$\implies \left(2^{-2n}\binom{2n}{n}\right)^2 \sim \frac{1}{\pi n}$$

■

Proposition 1.7 (Countable subadditivity)

Let (A_n) be a collection of events in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n\right) \leq \sum \mathbb{P}(A_n)$$

Proof. Let $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. By definition, it's immediate that (B_n) is a disjoint collection of sets in $\mathcal{F}$, and $\bigcup B_n = \bigcup A_n$, so

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n\right) = \mathbb{P}\left(\bigcup B_n\right) &= \sum \mathbb{P}(B_n) \\ &\leq \sum \mathbb{P}(A_n) \end{aligned}$$

since $B_n \subseteq A_n$ for all n. ■

Proposition 1.8 (Continuity of probability measures)

Take a sequence of events $(A_n)_{n\in\mathbb{N}}$ such that $A_n \subseteq A_{n+1}$ for all n – we say (A_n) is an *increasing* sequence.

Then $\mathbb{P}(A_n)$ is increasing and converges. In particular,

$$\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcup A_n\right)$$

as $n \to \infty$.

Proof. Define $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_n)$. Then

$$\bigcup_{k=1}^{n} B_k = A_n$$

and the B_n's are disjoint.Then

$$\begin{aligned} \mathbb{P}(A_n) &= \mathbb{P}\left(\bigcup_{k=1}^{n} B_k\right) \\ &= \sum_{k=1}^{n} \mathbb{P}(B_k) \to \sum_{k=1}^{\infty} \mathbb{P}(B_k) \end{aligned}$$

Also,

$$\begin{aligned} \sum_{k=1}^{\infty} \mathbb{P}(B_k) &= \mathbb{P}\left(\bigcup B_k\right) \\ &= \mathbb{P}\left(A_k\right) \end{aligned}$$

■

Corollary 1.9

Analogously, take a sequence of events (A_n) such that $A_{n+1} \subseteq A_n$ for all n – we say (A_n) is a decreasing sequence. Then

$$\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcap A_n\right)$$

Proof. Take complements and use the previous claim. ■

Proposition 1.10 (Inclusion-exclusion formula for probability measures)

Let $A_1, \dots, A_n \in \mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

Proof. We proceed by induction.

Base case. ADD

Inductive step. We have

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n\right) \\
&= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i \cap A_n\right) \\
&= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right) \\
&= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)
\end{aligned}$$

Applying the inductive hypothesis, we get

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) \\
\mathbb{P}\left(\bigcup_{i=1}^{n-1} B_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k} \cap A_n)
\end{aligned}$$

and subbing this in, we will find that everything works out. ■

Proposition 1.11 (Bonferroni inequalities)

Let $A_1, \dots, A_n$ be events. Then

$$\mathbb{P}\left(\bigcup A_i\right) \begin{cases} \leq \sum_{k=1}^{r} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ odd} \\ \geq \sum_{k=1}^{r} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ even} \end{cases}$$

where those sums are just truncated expansions of inclusion-exclusion.

Proof. We induct again.

Base case. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$

Inductive step. Suppose r is odd. Then

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)$$

By the inductive hypotheses,

$$\mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) \leq \sum_{k=1}^{r} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$

$$\mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right) \geq \sum_{k=1}^{r-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n)$$

where in the former, we choose to apply the identity for odd r, and in the latter, we apply it for even $r-1$. Substituting much as for the proof of the inclusion-exclusion formula, we obtain the result. The even case is analogous. ∎

Let us now consider the special case where Ω is a finite set and $\mathbb{P}(A) = \mid A \mid / \mid \Omega \mid$. Then we have

Corollary 1.12 (Inclusion-exclusion principle for sets)

Suppose Ω is finite. Then

$$\mid A_1 \cup \dots \cup A_n \mid = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mid A_{i_1} \cap \dots \cap A_{i_n} \mid$$

Proof. Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathbb{P}(A) = \mid A \mid / \mid \Omega \mid$. Then, apply inclusion-exclusion and cancel $1/\mid \Omega \mid$ from both sides. ∎

§2 Basic applications

§2.1 Counting surjections

Define a probability space as follows:

$$\Omega = \{f : \{1, \dots, n\} \to \{1, \dots, m\}\}$$

$$A = \{f \in \Omega : f \text{ is a surjection}\}$$

What is $\mid A \mid$? Let us define a set of events

$$A_i = \{f \in \Omega : i \notin \{f(1), \dots, f(n)\}\},\ i \in \{1, \dots, m\}$$

so that

$$\begin{aligned}
A &= A_1^{\mathsf{c}} \cap \cdots \cap A_m^{\mathsf{c}} = (A_1 \cup \cdots \cup A_m)^{\mathsf{c}} \\
\implies \mid A \mid &= \mid \Omega \mid - \mid A_1 \cup \cdots \cup A_m \mid \\
&= \mid \Omega \mid - \sum_{k=1}^{m} (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le m} \mid A_{i_1} \cap \cdots \cap A_{i_k} \mid \\
&= \underbrace{\mid \Omega \mid}_{m^n} - \sum_{k=1}^{m} (-1)^{k+1} \binom{m}{k} (m-k)^n \\
&= \sum_{k=0}^{m} (-1)^k \binom{m}{k} (m-k)^n
\end{aligned}$$

Lecture 4 – 30/01/26

§2.2 Counting derangements

Definition 2.1. A *derangement* is a permutation with no fixed points.

Let

$$\begin{aligned}
\Omega &= \{\text{permutations of } \{1, \dots, n\}\} \\
A &= \{\text{derangements}\}
\end{aligned}$$

Let

$$A_i = \{f \in \Omega : f(i) = i\}$$

so that

$$\begin{aligned}
&A = A_1^{\mathsf{c}} \cap \cdots \cap A_n^{\mathsf{c}} = (A_1 \cup \cdots \cup A_n)^{\mathsf{c}} \\
&\implies \mathbb{P}(A) = 1 - \mathbb{P}\left(\bigcup A_i\right) = 1 - \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le n} \underbrace{\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})}_{\frac{(n-k)!}{n!}} \\
&\qquad = 1 - \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} \\
&\qquad = 1 - \sum_{k=1}^{n} (-1)^{k+1} \frac{1}{k!} \\
&\qquad = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \to e^{-1}
\end{aligned}$$

§3 Independence and conditional probability

Definition 3.1 (Independence)**.** Let A, $B \in \mathcal{F}$. We say A and B are *independent* and write $A \perp B$ if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Likewise, a countable collection of events $(A_n)_{n\in\mathbb{N}}$ is said to be independent if, for distinct $i_1, \ldots, i_k$,

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \prod_{j=1}^{k} \mathbb{P}(A_{i_j})$$

> **Remark.** Pairwise independence of events does not imply independence.

Example

Toss a fair coin twice. We have

$$\begin{aligned} \Omega &= \{(0,0),(0,1),(1,0),(1,1)\} \\ \mathbb{P}(\{\omega\}) &= \frac{1}{4} \end{aligned}$$

Let

$$\begin{aligned} A &= \{(0,0),(0,1)\} \\ B &= \{(0,0),(1,0)\} \\ C &= \{(1,0),(0,1)\} \end{aligned}$$

We have $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C)$, and furthermore we can check that $A \perp B$, $A \perp C$ and $B \perp C$. But
$P(A \cap B \cap C) = \mathbb{P}(\emptyset) = 0$, so they are not independent.

Proposition 3.2

If A is independent of B, then A is independent of B^{c}.

Proof. We have

$$\begin{aligned} \mathbb{P}(A \cap B^{\mathsf{c}}) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A)(1 - \mathbb{P}(B)) \\ &= \mathbb{P}(A)\mathbb{P}(B^{\mathsf{c}}) \end{aligned}$$

■

Definition 3.3 (Conditional probability)**.** Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. For $A \in \mathcal{F}$ we write the *conditional probability of A given B*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Remark. Notably, if $A \perp B$, then $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, as we would expect.

Proposition 3.4 (Countable additivity for conditional probability)
Suppose $(A_n)_{n\in\mathbb{N}}$ is a disjoint sequence in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n \mid B\right) = \sum \mathbb{P}(A_n \mid B)$$

Proof. We have

$$\begin{aligned}
\mathbb{P}\left(\bigcup A_n \mid B\right) &= \frac{\mathbb{P}\left(\bigcup A_n \cap B\right)}{\mathbb{P}(B)} \\
&= \frac{\mathbb{P}\left(\bigcup (A_n \cap B)\right)}{\mathbb{P}(B)} \\
&= \sum \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} \\
&= \sum \mathbb{P}(A_n \mid B)
\end{aligned}$$

■

Proposition 3.5 (Law of total probability)
Suppose $(B_n)_{n\in\mathbb{N}}$ is a disjoint collection in $\mathcal{F}$, $\bigcup B_n = \Omega$, and $\mathbb{P}(B_n) > 0$ for every n. Let $A \in \mathcal{F}$, then

$$\mathbb{P}(A) = \sum_n \mathbb{P}(A \mid B_n)\mathbb{P}(B_n)$$

Proof. We have

$$\begin{aligned}
\mathbb{P}(A) &= \mathbb{P}\left(A \cap \left(\bigcup B_n\right)\right) \\
&= \mathbb{P}\left(\bigcup (B_n \cap A)\right) \\
&= \sum \mathbb{P}(B_n \cap A) \\
&= \sum \mathbb{P}(A \mid B_n)\mathbb{P}(B_n)
\end{aligned}$$

■

Proposition 3.6 (Bayes' theorem)
Suppose (B_n) are disjoint, $\bigcup B_n = \Omega$ and $P(B_n) > 0$ for every n. Let $A \in \mathcal{F}$, then

$$\mathbb{P}(B_n \mid A) = \frac{\mathbb{P}(A \mid B_n) \cdot \mathbb{P}(B_n)}{\sum_k \mathbb{P}(A \mid B_k)\mathbb{P}(B_k)}$$

Proof sketch. Use the definitions of $\mathbb{P}(B_n \mid A)$ and $\mathbb{P}(A \mid B_n)$, and expand the bottom using the law of total probability. ■

This formula is the basis of so-called Bayesian statistics. We know the probability of $\mathbb{P}(B_n)$ and we have a model which gives us $\mathbb{P}(A \mid B_n)$. With the formula, we find the *posterior* probability of B_n given that A occurred.

Example (False positives for a rare condition)

Suppose that condition A affects 0.1% of the population. We have a medical test which is positive for 98% of the affected population and 1% of the unaffected population. What is the probability that a random individual who tests positive has the disease?

Put $A = \{\text{individual suffers from } A\}$, $P = \{\text{individual tests positive}\}$. We have $\mathbb{P}(A) = 0.001$, $\mathbb{P}(P \mid A) = 0.98$, $\mathbb{P}(P \mid A^{\mathsf{c}}) = 0.01$, so

$$\begin{aligned}\mathbb{P}(A \mid P) &= \frac{\mathbb{P}(P \mid A)\mathbb{P}(A)}{\mathbb{P}(P \mid A)\mathbb{P}(A) + \mathbb{P}(P \mid A^{\mathsf{c}})\mathbb{P}(A^{\mathsf{c}})} \\ &= 0.089\end{aligned}$$

In particular, we can see

$$\mathbb{P}(A \mid P) = \frac{1}{1 + \frac{\mathbb{P}(P|A^{\mathsf{c}})\mathbb{P}(A^{\mathsf{c}})}{\mathbb{P}(P|A)\mathbb{P}(A)}}$$

Since $\mathbb{P}(A^{\mathsf{c}})$ and $\mathbb{P}(P \mid A)$ are both very close to one, the relevant ratio is

$$\frac{\mathbb{P}(P \mid A^{\mathsf{c}})}{\mathbb{P}(A)}$$

But since $\mathbb{P}(P \mid A^{\mathsf{c}}) \gg \mathbb{P}(A)$, this causes a very small value for $\mathbb{P}(A \mid P)$.

Lecture n

Example
- What is the probability that I have two boys given that I have two children one of whom is a boy?

  Since no further information is given, we take all outcomes to be equally likely. Let

  $$\mathrm{BG} = \{\text{elder child a boy, younger a girl}\}$$

  with GB, BB, GG defined analogously. Then we want

  $$\begin{aligned}\mathbb{P}(\mathrm{BB} \mid \mathrm{BB} \cup \mathrm{BG} \cup \mathrm{GB}) &= \frac{\mathbb{P}(\mathrm{BB})}{\mathbb{P}(\mathrm{BB} \cup \mathrm{BG} \cup \mathrm{GB})} \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}\end{aligned}$$

- What about $\mathbb{P}(\text{both boys} \mid \text{eldest a boy})$? This is

  $$\mathbb{P}(\mathrm{BB} \mid \mathrm{BB} \cup \mathrm{BG}) = \frac{1}{2}$$

- What about $\mathbb{P}(\text{both boys} \mid \text{one a boy born on a Thursday})$? Denote a boy born on Thursday by T, a boy not born on Thursday by N, and a girl by G. Then we want

  $$\begin{aligned}&\mathbb{P}(\mathrm{TT} \cup \mathrm{TN} \cup \mathrm{NT} \mid \mathrm{TT} \cup \mathrm{TG} \cup \mathrm{GT} \cup \mathrm{TN} \cup \mathrm{NT}) \\ &= \frac{\mathbb{P}(\mathrm{TT} \cup \mathrm{TN} \cup \mathrm{NT})}{\mathbb{P}(\mathrm{TT} \cup \mathrm{TG} \cup \mathrm{GT} \cup \mathrm{TN} \cup \mathrm{NT})} \\ &= \frac{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right)\left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right)\left(\frac{1}{2} \cdot \frac{1}{7}\right)}{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \cdot \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{1}{2} \cdot \frac{1}{7}\right) + \left(\frac{1}{2} \cdot \frac{1}{7}\right)\left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right)\left(\frac{1}{2} \cdot \frac{1}{7}\right)} \\ &=; \frac{13}{27}\end{aligned}$$

  Morally speaking, we're providing information that helps specify the boy in question more, which brings the probability closer to a half.

Example (Simpson's paradox)

Consider the following data:

All applicants	Admitted	Rejected	% Admitted
State	25	25	50%
Indep.	28	22	56%

London schools	Admitted	Rejected	% Admitted
State	15	22	41%
Indep.	5	8	38%

Cambridge schools	Admitted	Rejected	% Admitted
State	10	3	77%
Indep.	23	14	62%

– paradoxically, the state admission rate is higher when the school locations are taken individually, but the independent admission right is higher for the combined data. This phenomenon is called *confounding*, and arises when we aggregate data from disparate populations.

Let

$$\begin{aligned} A &= \{\text{individual is admitted}\} \\ B &= \{\text{individual from London}\} \\ C &= \{\text{individual from state school}\} \end{aligned}$$

Then

$$\begin{aligned} \mathbb{P}(A \mid B \cap C) &> \mathbb{P}(A \mid B \cap C^{\mathsf{c}}) \\ \mathbb{P}(A \mid B^{\mathsf{c}} \cap C) &> \mathbb{P}(A \mid B^{\mathsf{c}} \cap C^{\mathsf{c}}) \\ \text{but } \mathbb{P}(A \mid C) &< \mathbb{P}(A \mid C^{\mathsf{c}}) \end{aligned}$$

Let's examine how this can happen. We have

$$\begin{aligned} \mathbb{P}(A \mid C) &= \mathbb{P}(A \cap B \mid C) + \mathbb{P}(A \cap B^{\mathsf{c}} \mid C) \\ &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \cap B^{\mathsf{c}} \cap C)}{\mathbb{P}(C)} \\ &= \frac{\mathbb{P}(A \mid B \cap C)\mathbb{P}(B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \mid B^{\mathsf{c}} \cap C)\mathbb{P}(B^{\mathsf{c}} \cap C)}{\mathbb{P}(C)} \\ &= \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^{\mathsf{c}} \cap C^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}} \mid C^{\mathsf{c}}) \\ &> P(A \mid B \cap C^{\mathsf{c}})\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^{\mathsf{c}} \cap C^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}} \mid C) \end{aligned}$$

by the first two inequalities from above. Now, if $\mathbb{P}(B \mid C) = \mathbb{P}(B \mid C^{\mathsf{c}})$, then we would get $\mathbb{P}(A \mid C) > \mathbb{P}(A \mid C^{\mathsf{c}})$, which is the intuitive expectation – but here this does not hold, and so we can make the 'paradox' hold.

Even more simply, this is just the fact that, if $\frac{a_1}{a_2} > \frac{b_1}{b_2}$ and $\frac{c_1}{c_2} > \frac{d_1}{d_2}$, it does not necessarily hold that $\frac{a_1+c_1}{a_2+c_2} > \frac{b_1+d_1}{b_2+d_2}$.

§4 Probability distributions

§4.1 Discrete probability distributions

Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with Ω either finite or countable – that is, $\Omega = \{\omega_1, \omega_2, \dots\}$. As normal, we take $\mathcal{F} = \mathcal{P}(\Omega)$, and so we need only care about $\mathbb{P}(\{\omega_i\})$ for every i.

Definition 4.1 (Discrete probability distributions)**.** For such a setup, we call the sequence of probabilities $(\mathbb{P}(\{\omega_i\}))_i$ a *discrete probability distribution*. We may also write $p_i = \mathbb{P}(\{\omega_i\})$.

Let us consider some examples of discrete probability distributions.

① *Bernoulli distribution*. The Bernoulli distribution models the outcome of a toss of a biased coin. We have $\Omega = \{0, 1\}$, $p_0 = \mathbb{P}(\{0\}) = 1 - p$, $p_1 = \mathbb{P}(\{1\}) = p$ for some p with $p \in [0, 1]$.

② *Binomial distribution*. We denote this $\mathrm{Bin}(n, p)$, with $n \in \mathbb{N}$, $p \in [0, 1]$. It models the number of heads seen for n independent tosses of a p-biased coin. We have

$$\begin{aligned} \Omega &= \{0, 1, \dots, n\} \\ p_k &= \binom{n}{k} p^k (1-p)^{n-k} \end{aligned}$$

③ *Multinomial distribution*. We denote this $\mathrm{M}(n, p_1, \dots, p_k)$, where $p_i \geq 0$, $\sum p_i = 1$. Given k boxes and n balls placed in box i with p_i, the multinomial distribution gives the probability of n_l balls in box l for every n. We have

$$\begin{aligned} \Omega &= \left\{ (n_1, \dots, n_k) \in \mathbb{N}^k : \sum_{i=1}^{k} n_i = n \right\} \\ \mathbb{P}(n_1, \dots, n_k) &= \binom{n}{n_1} n_1^{p_1} \cdot \binom{n - n_1}{n_2} p_2^{n_2} \cdots \binom{n - n_1 - \dots - n_{k-1}}{n_k} p_k^{n_k} \\ &= \binom{n}{n_1, n_2, \dots, n_k} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k} \end{aligned}$$

④ *Geometric distribution*. The geometric distribution gives the probaiity that the first heads achieved on a p-coin is after k tosses:

$$\begin{aligned} \Omega &= \{1, 2, \dots\} \\ p_k &= (1-p)^{k-1} p \end{aligned}$$

Alternatively, we might want to model the probability that we see k tails before the first heads:

$$\begin{aligned} \Omega &= \{0, 1, \dots\} \\ p'_k &= (1-p)^k p \end{aligned}$$

⑤ *Poisson distribution*. Suppose customers arrive into a shop in $[0, 1]$. Discretise $[0, 1]$ into $\left\{ \left[\frac{i-1}{N}, \frac{i}{N} \right] \right\}, i = 1, \dots, N$, $N \in \mathbb{N}$. Independently, in each interval, a customer arrives with probability p, and doesn't arrive with probability $1 - p$.

Then

$$\mathbb{P}(k \text{ customers have arrived}) = \binom{N}{k} p^k (1-p)^{N-k}$$

– in other words, this follows a binomial distribution. Now, take $p = \frac{\lambda}{N}$, so

$$\begin{aligned}
\mathbb{P}(k \text{ customers have arrived}) &= \frac{N!}{k!(N-k)!}\left(\frac{\lambda}{N}\right)^k \left(1-\frac{\lambda}{N}\right)^{N-k} \\
&= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{N(N-1)\dots(N-k+1)}{N^k}}_{\to 1} \underbrace{\left(1-\frac{\lambda}{N}\right)^{N-k}}_{\to e^{-\lambda}} \\
&\to e^{-\lambda}\frac{\lambda^k}{k!}
\end{aligned}$$

as $N \to \infty$. Hence we have

$$\begin{aligned}
\Omega &= \{0, 1, 2, \dots\} \\
p_k &= e^{-\lambda}\frac{\lambda^k}{k!}
\end{aligned}$$

and we denote this $\mathrm{Poi}(\lambda)$.

§4.2 Random variables

Definition 4.2. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \to \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \le x\} \in \mathcal{F}$$

for any x. We have the shorthand notation $\{X \le x\}$ to refer to $\{\omega \in \Omega : X(\omega) \le x\}$, and $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.

Definition 4.3. Let $A \in \mathcal{F}$. Define the *indicator* of A

$$1_A(\omega) = 1(\omega \in A) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

Since $A \in \mathcal{F}$, 1_A is a random variable.

Definition 4.4. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned}
F_X(x) &: \mathbb{R} \to [0,1] \\
&x \mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \le x\}) = \mathbb{P}(X \le x)
\end{aligned}$$

Definition 4.5. $X = (X_1, \dots, X_n)$ is called a *random variable in* $\mathbb{R}^n$ if

$$(X_1, \dots, X_n) : \Omega \to \mathbb{R}^n$$

is a function such that, for any $x_1, \dots, x_n$,

$$\{\omega : X_1(\omega) \le x_1, \dots, X_n(\omega) \le x_n\} \in \mathcal{F}$$

which we denote with the shorthand $\{X_1 \le x_1, \dots, X_n \le x_n\}$.

Note that $(X_1, \dots, X_n)$ is a random variable if and only if all of $X_1, \dots, X_n$ are.

Definition 4.6. A random variable X is called *discrete* if it takes values in some countable set S.

Definition 4.7. For a discrete random variable, we are able to write

$$p_x = \mathbb{P}(X = x) = \mathbb{P}(\omega : X(\omega) = x)$$

Then, $(p_x)_{x \in S}$ is called the *probability mass function* of X.

Now we are able to say things like

$$\text{for any } A \subseteq S, \mathbb{P}(X \in A) = \sum_{x \in A} p_x$$

If p_x follows a given distribution, then we say X is a random variable with that distribution or a random variable following that distribution.

Definition 4.8. Let $X_1, \dots, X_n$ be discrete random variables with values in $S_1, \dots, S_n$. They are called *independent* if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \dots \mathbb{P}(X_n = x_n)$$

for all $x_1 \in S_1, \dots, x_n \in S_n$.

Example 4.9

Toss a p-coin N times independently. Then

$$\Omega = \{0, 1\}^N$$

and every $\omega \in \Omega$ is of the form $(\omega_1, \dots, \omega_N)$, where ω_i is the i^{th} outcome. Hence

$$p_\omega = \mathbb{P}(\{\omega\}) = \prod_{k=1}^{N} p^{\omega_k} (1-p)^{1-\omega_k}$$

Let us also define random variables X_k for each k, satisfying $X_k(\omega) = \omega_k$. Clearly X_k is a Bernoulli r.v. of parameter p. We claim $X_1, \dots, X_N$ are independent: we have

$$\begin{aligned} P(X_1 = x_1, \dots, X_N = x_N) &= \prod_{k=1}^{N} p^{x_\omega} (1-p)^{1-x_\omega} \\ &= \prod_{k=1}^{N} \mathbb{P}(X_i = x_i) \end{aligned}$$

Lecture 7 – 06/02/26

§4.3 Expectation

Definition 4.10. Suppose Ω is finite or countable, and let $X : \Omega \to \mathbb{R}$ be a discrete random variable. We say X is *non-negative* if $X \geq 0$ – then we define the *expectation* of X by

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) \cdot X(\omega)$$

We can equivalently let

$$\Omega_X = \{X(\omega) : \omega \in \Omega\}$$

so that

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}(\{\omega\}) \\
&= \sum_{x \in \Omega_X} \sum_{\omega \in \{X = x\}} X(\omega)\mathbb{P}(\{\omega\}) \\
&= \sum_{x \in \Omega_X} \sum_{\omega \in \{X = x\}} x\mathbb{P}(\{\omega\}) \\
&= \sum_{x \in \Omega_X} x \sum_{\omega \in \{X = x\}} \mathbb{P}(\{\omega\}) \\
&= \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x)
\end{aligned}$$

or in other words the average of the values taken by X, weighted by their probabilities.

Example 4.11

Suppose $X \sim \text{Bin}(n, p)$. Then

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{k=0}^{N} k\mathbb{P}(X = k) \\
&= \sum_{k=0}^{N} k \binom{N}{k} p^k (1-p)^{N-k} \\
&= \sum_{k=0}^{N} k \frac{N!}{(N-k)!k!} p^k (1-p)^{N-k} \\
&= \sum_{k=0}^{N-1} N \cdot \binom{N-1}{k} p^{k-1} (1-p)^{N-k} \cdot p \\
&= Np \cdot (p + 1 - p)^{N-1} = Np
\end{aligned}$$

Example 4.12

Suppose $X \sim \text{Poi}(\lambda)$. Then

$$\begin{aligned}\mathbb{E}[X] &= \sum_{k=0}^{\infty} k\mathbb{P}(X = k) \\ &= \sum_{k=0}^{\infty} ke^{-\lambda}\frac{\lambda^k}{k!} \\ &= e^{-\lambda} \cdot \lambda \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda\end{aligned}$$

Let X be a random variable – then we define

$$\begin{aligned}X^+ &= \max(X, 0) \\ X^- &= \max(-X, 0)\end{aligned}$$

so $X = X^+ - X^-$ and $|X| = X^+ + X^-$. Supposing X is discrete, we can define the expectations of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$, since they are both non-negative.

If at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, we define

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$$

If both are infinite, the expectation of X is undefined.

Also, if $\mathbb{E}[|X|]$ (which we define in the obvious way) is $< \infty$, we say X is integrable. When $\mathbb{E}[X]$ is well-defined, we have

$$\mathbb{E}[X] = \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x)$$

We have some properties of expectations as follows:

- If $X \geq 0$, then $\mathbb{E}[X] \geq 0$.
- If $X \geq 0$ and $\mathbb{E}[X] = 0$, then $\mathbb{P}(X = 0) = 1$.
- If $c \in \mathbb{R}$, then $\mathbb{E}[cX] = c\mathbb{E}[X]$ and $\mathbb{E}[c + X] = c + \mathbb{E}[X]$.
- If X and Y are random variables (and X, Y integrable), then $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
- In the countable case, we take $c_1, \dots, c_n \in \mathbb{R}$ and $X_1, \dots, X_n$ integrable, and then we have

$$\mathbb{E}\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i \mathbb{E}[X_i]$$

Let us give a proof of the final identity in the case where Ω is countable. We have

$$\begin{aligned}\mathbb{E}\left[\sum_n X_n\right] &= \sum_{\omega \in \Omega}\left(\sum_n X_n(\omega)\right)\mathbb{P}(\{\omega\}) \\ &= \sum_n \sum_{\omega \in \Omega} X_n(\omega)\mathbb{P}(\{\omega\})\end{aligned}$$

$$= \sum_n \mathbb{E}[X_n]$$

where we can interchange sums by the assumption that the random variables are integrable.

We also have the useful result that, if $X = 1(A)$ for some event A,

$$\mathbb{E}[X] = \mathbb{P}(A)$$

Proposition 4.13

Let $g : \mathbb{R} \to \mathbb{R}$, and define the random variable $g(X)$ by

$$g(X)(\omega) = g(X(\omega))$$

Then

$$\mathbb{E}[g(X)] = \sum_{x \in \Omega_X} g(x)\mathbb{P}(X = x)$$

Proof. Put $Y = g(X)$, then

$$\begin{aligned}
\{Y = y\} &= \{\omega : Y(\omega) = y\} \\
&= \{\omega : g(X(\omega)) = y\} \\
&= \{\omega : X(\omega) \in g^{-1}(\{y\})\} \\
&= \{X \in g^{-1}(\{y\})\}
\end{aligned}$$

Hence

$$\begin{aligned}
\mathbb{E}[Y] &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(Y = y) \\
&= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(X \in g^{-1}(\{y\})) \\
&= \sum_{y \in \Omega_Y} y \sum_{x \in g^{-1}(\{y\})} \mathbb{P}(X = x) \\
&= \sum_{y \in \Omega_Y} \sum_{x \in g^{-1}(\{y\})} g(x)\mathbb{P}(X = x) \\
&= \sum_{x \in \Omega_X} g(x) \cdot \mathbb{P}(X = x)
\end{aligned}$$

■

Proposition 4.14

Suppose that $X \geq 0$ and takes integer values. Then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \sum_{k=0}^{\infty} \mathbb{P}(X > k)$$

Proof. Since $X \geq 0$ and $X \in \mathbb{N}$, we have

$$X = \sum_{k=1}^{\infty} 1(X \geq k) = \sum_{k=0}^{\infty} 1(X > k)$$

Taking expectations, we obtain

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}\left[\sum_{k=1}^{\infty} 1(X \geq k)\right] & &= \mathbb{E}\left[\sum_{k=0}^{\infty} 1(X > k)\right] \\ &= \sum_{k=1}^{\infty} \mathbb{E}[1(X \geq k)] & &= \sum_{k=0}^{\infty} \mathbb{E}[1(X > k)] \\ &= \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) & &= \sum_{k=0}^{\infty} \mathbb{P}(X > k) \end{aligned}$$

■

Lecture $n+1$ – 09/02/26

We present another proof of the inclusion-exclusion formula. First, we should establish some properties about indicator functions:

- $1(A^c) = 1 - 1(A)$;
- $1(A \cap B) = 1(A) \cdot 1(B)$;
- $1(A \cup B) = 1 - (1 - 1(A))(1 - 1(B))$.

More generally,

$$\begin{aligned} 1(A_1 \cup \cdots \cup A_n) &= 1 - \prod_{i=1}^{n} (1 - 1(A_i)) \\ &= \sum_{i=1}^{n} 1(A_i) - \sum_{1 \leq i_1 < i_2 \leq n} 1(A_{i_1} \cap A_{i_2}) + \cdots + (-1)^{n+1} 1(A_1 \cap \cdots \cap A_n) \end{aligned}$$

Then, we can take expectations of both sides and apply $\mathbb{E}[1(A)] = \mathbb{P}(A)$ to get

$$\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

§4.4 Moments, variance and covariance

Definition 4.15. Let X be a random variable and $r \in \mathbb{N}$. We call $\mathbb{E}[X^r]$, as long as it is well-defined, the r^{th} *moment* of X.

Definition 4.16. The *variance* of X is given by

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Furthermore, the *standard deviation* of X is given by $\sqrt{\mathrm{Var}(X)}$.

We have some properties of variance as follows:

- $\operatorname{Var}(X) \geq 0$, since it is the expectation of a squared quantity;
- If $\operatorname{Var}(X) = 0$, then $\mathbb{P}(X = \mathbb{E}[X]) = 1$;
- If $c \in \mathbb{R}$, then $\operatorname{Var}(cX) = c^2 \operatorname{Var}(X)$ and $\operatorname{Var}(X + c) = \operatorname{Var}(X)$.

Lemma 4.17

The variance of X is also given by

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Proof. We have

$$\begin{aligned}
\operatorname{Var}(X) &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\
&= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\
&= \mathbb{E}[X^2] - \mathbb{E}[X]^2
\end{aligned}$$

■

Lemma 4.18

The function $\mathbb{E}[(X-c)^2]$ is minimised when $c = \mathbb{E}[X]$, and at that point is equal to $\operatorname{Var}(X)$.

Proof. We have

$$\begin{aligned}
f(c) &= \mathbb{E}[(X-c)^2] \\
&= \mathbb{E}[X^2] - 2c\mathbb{E}[X] + c^2 \\
\implies f'(c) &= -2\mathbb{E}[X] + 2c = 0 \\
\implies c &= \mathbb{E}[X]
\end{aligned}$$

■

Example 4.19 ① Suppose $X \sim \text{Bin}(n, p)$. Recall $\mathbb{E}[X] = np$. We have

$$\text{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$$

Then

$$\begin{aligned}
\mathbb{E}[X(X-1)] &= \sum_{k=2}^{n} k(k-1)\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k} \\
&= (n-1)np^2 \sum_{k=0}^{n-2} \binom{n-2}{k} p^k (1-p)^{n-2-k} \\
&= (n-1)np^2
\end{aligned}$$

so that $\text{Var}(X) = (n-1)np^2 + np - n^2p^2 = np(1-p)$.

② Suppose $X \sim \text{Poi}(\lambda)$. Recall $\mathbb{E}[X] = \lambda$. We have

$$\begin{aligned}
\mathbb{E}[X(X-1)] &= \sum_{k=2}^{\infty} k(k-1)e^{-\lambda}\frac{\lambda^k}{k!} \\
&= e^{-\lambda}l^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2
\end{aligned}$$

So $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Definition 4.20. Let X and Y be random variables. Then the *covariance* of X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

We have some properties:

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
- $\text{Cov}(X, X) = \text{Var}(X)$;
- for any $c \in \mathbb{R}$, $\text{Cov}(cX, Y) = c\,\text{Cov}(X, Y)$ and $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$;
- for any $c \in \mathbb{R}$, $\text{Cov}(c, X) = 0$.

Lemma 4.21

The covariance of X and Y is given by

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Proof. Expand. ∎

Lemma 4.22

For random variables X and Y, we have

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$$

Proof. We have

$$\begin{aligned}\operatorname{Var}(X+Y) &= \mathbb{E}[((X-\mathbb{E}[X])+(Y-\mathbb{E}[Y]))^2] \\ &= \operatorname{Var}(X) + 2\operatorname{Cov}(X,Y) + \operatorname{Var}(Y)\end{aligned}$$

■

Lemma 4.23

For $c_1, \dots, c_n$ and $d_1, \dots, d_n \in \mathbb{R}$,

$$\operatorname{Cov}\left(\sum_{i=1}^n c_i X_i, \sum_{j=1}^n d_i Y_i\right) = \sum_{i,j=1}^n c_i d_j \operatorname{Cov}(X_i, Y_j)$$

Corollary 4.24

We have

$$\begin{aligned}\operatorname{Var}\left(\sum_{i=1}^n X_i\right) &= \operatorname{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \operatorname{Var}(X_i) + \sum_{i\neq j} \operatorname{Cov}(X_i, X_j)\end{aligned}$$

Let us consider some results on independence.

Lemma 4.25

Suppose X_1, X_2 and X_3 are independent – then X_1 and X_2 are independent.

Proof. We have

$$\begin{aligned}\mathbb{P}(X_1 = x_1, X_2 = x_2) &= \sum_{x_3} \mathbb{P}(X_1 = x_1, X_2 = x_2, X_3 = x_3) \\ &= \sum_{x_3} \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2)\mathbb{P}(X_3 = x_3) \\ &= \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2)\end{aligned}$$

■

> **Remark.** The result obviously generalises to any subset of a set of independent events being independent.

Lemma 4.26

Let X and Y be two independent random variables, and let f, $g : \mathbb{R} \to \mathbb{R}_+$. Then

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$$

Proof. Let the random variable $Z = (X, Y)$, so $h(Z) = f(X)g(Y)$. Then

$$\begin{aligned}
\mathbb{E}[h(Z)] &= \sum_{(x,y)} h(x,y)\mathbb{P}(Z = (x,y)) \\
&= \sum_{(x,y)} f(x)g(y)\mathbb{P}(X = x, Y = y) \\
&= \sum_{(x,y)} f(x)g(y)\mathbb{P}(X = x)\mathbb{P}(Y = y) \\
&= \mathbb{E}[f(X)] \cdot \mathbb{E}[g(Y)]
\end{aligned}$$

■

Proposition 4.27

If X and Y are independent, $\mathrm{Cov}(X, Y) = 0$.

Remark. The converse is not true.

Example 4.28

Let X_1, X_2, X_3 be independent Bernoulli random variables with $p = \frac{1}{2}$. Let $Y_1 = 2X_1 - 1$, $Y_2 = 2X_2 - 1$. Also let $Z_1 = Y_1X_3$, $Z_2 = Y_2X_3$. Then $\mathbb{E}[Y_1] = \mathbb{E}[Y_2] = 0$, and $\mathbb{E}[Z_1] = \mathbb{E}[Z_2] = 0$ by independence. Then

$$\mathrm{Cov}(Z_1, Z_2) = \mathbb{E}[Z_1Z_2] = \mathbb{E}[Y_1Y_2X_3^2]$$

But we claim Z_1 and Z_2 are not independent. Observe $\mathbb{P}(Z_1 = 0, Z_2 = 0) = \mathbb{P}(X_3 = 0) = \frac{1}{2}$. But $\mathbb{P}(Z_1 = 0) = \mathbb{P}(Z_2 = 0) = \frac{1}{2}$.

§4.5 Inequalities on random variables

Proposition 4.29 (Markov's inequality)

Let X be a non-negative random variable and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof. We have

$$\begin{aligned}
& X \geq a \cdot 1(X \geq a) \\
\implies & E[X] \geq a \cdot \mathbb{P}(X \geq a)
\end{aligned}$$

by taking expectations. ■

Proposition 4.30 (Chebyshev's inequality)
Let X be a random variable with $\mathbb{E}[X] < \infty$. Then, for any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof. Observe

$$\begin{aligned}
\mathbb{P}(|X - \mathbb{E}[X]| \geq a) &= \mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2) \\
&\leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{a^2} && \text{by Markov} \\
&= \frac{\text{Var}(X)}{a^2}
\end{aligned}$$

■

Proposition 4.31 (Cauchy-Schwarz)
Let X and Y be random variables, then

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]}$$

Proof. Assume that $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$ are finite, otherwise there is nothing to prove. Then, $|XY| \leq \frac{1}{2}(X^2 + Y^2)$, so, taking expectations, $\mathbb{E}[|XY|]$ is also finite. We may also assume $\mathbb{E}[X^2] > 0$, $\mathbb{E}[Y^2] > 0$. Let us also assume for the time being that $X, Y \geq 0$.

Let $t \in \mathbb{R}$ and consider $(X - tY)^2$. We have

$$\begin{aligned}
0 &\leq \mathbb{E}[(X - tY)^2] \\
&= \mathbb{E}[X^2] - 2t\mathbb{E}[XY] + t^2\mathbb{E}[Y^2] := f(t)
\end{aligned}$$

Then $f'(t) = -2\mathbb{E}[XY] + 2t\mathbb{E}[Y^2]$, so $\min f(t)$ is achieved at $t^* = \frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}$. We hence have

$$\begin{aligned}
0 &\leq \mathbb{E}[X^2] - 2\frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} + \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} \\
\implies \mathbb{E}[XY]^2 &\leq \mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]
\end{aligned}$$

■

Remark. We could also have just put in our value for $t*$ from the beginning – the minimisation approach is purely to provide rationale.

Remark. When is equality achieved? We would need $f(t^*) = 0$, that is

$$\begin{aligned}
&\mathbb{E}[(X - t^*Y)^2] = 0 \\
\implies &P(X = t^*Y) = \mathbb{P}(X - t^*Y = 0) = 1
\end{aligned}$$

so X is just a multiple of Y.

Definition 4.32 (Convexity)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is called *convex* if, for any $x, y \in \mathbb{R}$ and any $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

Proposition 4.33 (Jensen's inequality)

Let X be a random variable, and $X : \mathbb{R} \to \mathbb{R}$ a convex function. Then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

Remark. We can remember the direction of the inequality by putting $f(x) = x^2$ and using non-negativity of the variance.

Lemma 4.34

Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Then f is the supremum of all the lines lying below it. In other words, for all m in $\mathbb{R}$, there exist a, b in $\mathbb{R}$ such that $f(m) = am+b$ and $f(x) \geq ax + b$ for all x.

Proof of lemma. Let $m \in \mathbb{R}$, and suppose $x < m < y$. Then $m = tx + (1 - t)y$ for some $t \in [0, 1]$, or, equivalently,

$$t(m - x) = (1 - t)(y - m) \qquad (*)$$

By convexity, we have

$$\begin{aligned} f(m) = f(tx + (1 - t)y) &\leq tf(x) + (1 - t)f(y) \\ \implies t(f(m) - f(x)) &\leq (1 - t)(f(y) - f(m)) \\ (*) \implies \frac{f(m) - f(x)}{m - x} &\leq \frac{f(y) - f(m)}{y - m} \end{aligned}$$

Then set

$$a = \sup_{x<m} \frac{f(m) - f(x)}{m - x}$$

so that, for any $x < m < y$,

$$\frac{f(m) - f(x)}{m - x} \leq a \leq \frac{f(y) - f(m)}{y - m}$$

Hence, by rearrangement, we conclude $f(z) \geq a(z - m) + f(m)$, for any z (depending on whether z is greater than m or less than m, we use either the left or right inequality.) ∎

Proof of proposition. Let $m = \mathbb{E}[X]$. By the claim, we have $a, b \in \mathbb{R}$ such that $f(m) = am + b$ and $f(x) \geq ax + b$ for all x. We have

$$\begin{aligned} f(X) &\geq aX + b \\ \implies \mathbb{E}[f(X)] &\geq a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X]) \end{aligned}$$

∎

Remark. When is equality achieved?

Suppose f convex with the extra property that, for $n = \mathbb{E}[X]$, there exist a, $b \in \mathbb{R}$ with $f(m) = am + b$ but $f(x) > ax + b$, note the strict inequality, for $x \neq m$. When should X satisfy $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$?

Since $f(X) \geq aX + b$, we have $f(X) - (aX + b) \geq 0$. Taking expectations, we have

$$\mathbb{E}[f(X)] \geq a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X])$$

Since we want $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$, we must have

$$\begin{aligned} &\mathbb{E}[f(X) - (aX + b)] = 0 \\ \implies &\mathbb{P}(f(X) = aX + b) = 1 \end{aligned}$$

but then, since $f(x) > ax + b$ for $x \neq m$, we conclude $\mathbb{P}(X = m) = 1$ (if we didn't have that extra part, we'd conclude f is just a linear function).

Lecture $n + 1$

Proposition 4.35 (AM-GM inequality)

Let $x_1, \ldots, x_n > 0$. Then

$$\left(\prod_{k=1}^{n} x_k\right)^{\frac{1}{n}} \leq \frac{1}{n}\sum_{k=1}^{n} x_k$$

Lemma 4.36

Let f be a convex function. Then, for any $x_1, \ldots, x_n \in \mathbb{R}$ we have

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) \geq f\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right)$$

Proof of lemma. Let X be a random variable with $\mathbb{P}(X = x_i) = \frac{1}{n}$, $i = 1, \ldots, n$. Then

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) = \sum_{x} f(x)\mathbb{P}(X = x) = \mathbb{E}[f(X)]$$

so Jensen's gives

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) = f\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right)$$

as required. ■

Proof of proposition. Let $f(x) = -\log x$ and suppose $x_1, \ldots, x_n > 0$. Then

$$-\frac{1}{n}\sum_{k=1}^{n} \log x_k \geq -\log\left(\frac{1}{n}\sum_{k=1}^{n} x_n\right)$$

$$\implies \left(\prod_{k=1}^{n} x_k\right)^{\frac{1}{n}} \leq \frac{1}{n}\sum_{k=1}^{n} x_k$$

as required. ■

§4.6 Joint distribution and conditional expectation

Recall if X is a discrete random variable, then the disttribution of X is $(\mathbb{P}(X = x))_x$.

Definition 4.37. Let $X_1, \ldots, X_n$ be discrete random variables. Their *joint distribution* is defined to be the set of values of

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n)$$

for any $x_1, \ldots, x_n$.

> **Remark.** We can recover the distribution of any one of the variables from such a joint distribution with
>
> $$\mathbb{P}(X_1 = x_1) = \sum_{x_2,\ldots,x_n} \mathbb{P}(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$$
>
> In this context the distribution of the single variable is sometimes called its *marginal* distribution.

Definition 4.38. Let X and Y be discrete random variables. The *conditional distribution* of X given $Y = y$ is

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

By the law of probability we of course have

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x \mid Y = y)\mathbb{P}(Y = y)$$

Let X and Y be independent – what is the distribution of $X + Y$? We have

$$\begin{aligned}
\mathbb{P}(X + Y = z) &= \sum_y \mathbb{P}(X + Y = z, Y = y) \\
&= \sum_y \mathbb{P}(X = z - y, Y = y) \\
&= \sum_y \mathbb{P}(X = z - y)\mathbb{P}(Y = y)
\end{aligned}$$

where the last line is by independence. This is of course a convolution.

Example

Let $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$, with $\lambda, \mu > 0$ and $X \perp Y$. Then

$$\begin{aligned}\mathbb{P}(X+Y=n) &= \sum_{k=0}^{n} \mathbb{P}(X=n-k)\mathbb{P}(Y=k) \\ &= e^{-\lambda}\frac{\lambda^{n-k}}{(n-k)!} \cdot e^{-\mu}\frac{\mu^k}{k!} \\ &= \frac{e^{-(\lambda+\mu)}}{n!}\sum_{k=0}^{n}\binom{n}{k}\lambda^{n-k}\mu^k \\ &= e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^n}{n!}\end{aligned}$$

so in particular $X+Y \sim \text{Poi}(\lambda+\mu)$.

Let us derive conditional expectation for random variables. First, take $B \in \mathcal{F}$ to be an event with $\mathbb{P}(B) > 0$. We have

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X \cdot 1(B)]}{\mathbb{P}(B)}$$

and, if $X = 1(A)$, then

$$\mathbb{E}[1(A) \mid B] = \frac{\mathbb{E}[1(A) \cdot 1(B)]}{\mathbb{P}(B)} = \frac{\mathbb{E}[1(A \cap B)]}{\mathbb{P}(B)} = \mathbb{P}(A \mid B)$$

We can now formulate the *law of total expectation*.

Proposition 4.39 (Law of total expectation)

Let X be a random variable and (Ω_n) a partition of Ω with $\mathbb{P}(\Omega_n) > 0$ for any n. Then

$$\mathbb{E}[X] = \sum_n \mathbb{E}[X \mid \Omega_n]\mathbb{P}(\Omega_n)$$

Proof. Use indicators. We have

$$\mathbb{E}[X] = \mathbb{E}[X \cdot 1(\Omega)] = \mathbb{E}\left[X \cdot \sum_n 1(\Omega_n)\right] = \sum_n \mathbb{E}[X \mid \Omega_n]\mathbb{P}(\Omega_n)$$

■

Now, we want to condition expectation not on an event but on a particular random variable. We have

$$\begin{aligned}\mathbb{E}[X \mid Y=y] &= \frac{\mathbb{E}[X \cdot 1(Y=y)]}{\mathbb{P}(Y=y)} \\ &= \sum_x x \cdot \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)} \\ &= \sum_x x \cdot \mathbb{P}(X=x \mid Y=y)\end{aligned}$$

Let us now define $g(y) = \mathbb{E}[X \mid Y=y]$.

Definition 4.40. The expectation of X given Y, $\mathbb{E}[X \mid Y]$, is a random variable which is a function of Y. In particular,

$$\mathbb{E}[X \mid Y] = g(Y)$$

Example

Toss a p-coin independently n times, and let $X_i = 1(i^{\text{th}} \text{ toss is H})$. Let $Y_n = X_1 + \cdots + X_n$. Let us attempt to calculate $\mathbb{E}[X_1 \mid Y_n]$. First, let

$$\begin{aligned} g(y) = \mathbb{E}[X_1 \mid Y_n = y] &= \frac{\mathbb{E}[X_1 \cdot 1(Y_n = y)]}{\mathbb{P}(Y_n = y)} \\ &= \frac{\mathbb{P}(X_1 = 1, Y_n = y)}{\mathbb{P}(Y_n = y)} \\ &= \frac{p\binom{n-1}{y-1}p^{y-1}(1-p)^{n-y}}{\binom{n}{y}p^y(1-p)^{n-y}} \\ &= \frac{y}{n} \end{aligned}$$

Hence, $\mathbb{E}[X_1 \mid Y_n] = \frac{Y_n}{n}$.

We have some simple and uninspiring properties:

- $\mathbb{E}[cX \mid Y] = c\mathbb{E}[X \mid Y]$;
- $\mathbb{E}[c + X \mid Y] = c + \mathbb{E}[X \mid Y]$;
- $\mathbb{E}[c \mid Y] = c$;
- If $X_1, \ldots, X_n$ are random variables,

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i \mid Y\right] = \sum_{i=1}^{n} \mathbb{E}[X_i \mid Y]$$